USB CDC to DB25 8-bit Marker — Schematic v0.3-r2 (P0 polarity+TMR2+suspend — NOT fab) (NOT frozen for PCB fab)

Design pack v0.3 Adopt: regulated iso DC-DC | 6N137 VE(active-HIGH)+READY | 74HC14 invert-only | Busy NACK | Clear then Gap

GPIO map: PB0..PB7=MARKER_D0..D7 (atomic port); PB8=MARKER_READY; PA11=USB_DM; PA12=USB_DP; PA13=SWDIO; PA14=SWCLK; PH0/PH1=HSE 25MHz; PA15=ISO_PWR_EN (Suspend).

ERC note: MCU/USB-C/DB25 symbol pins are annotated by net-name labels; many pin_not_connected remain until pin-exact wires are finished in GUI. Opto channels below are hard-wired.

AUTO-WIRED key blocks (pin to label) review pass

=== A. USB Device / 3V3 / MCU / SWD (net-name annotated) ===

U1 net binding (provisional):

PB0=MARKER_D0 … PB7=MARKER_D7

PB8=MARKER_READY

PA11=USB_DM PA12=USB_DP

PA13=SWDIO PA14=SWCLK

PH0=OSC_IN PH1=OSC_OUT

NRST via R52 to +3V3; BOOT0 via R51 to GND

All VDD=+3V3; All VSS=GND

MCU PA15 = ISO_PWR_EN (active-high enable iso rail)

U14: VI=VBUS VO=+3V3 GND=GND (attach pins in GUI if ERC requires)

J1 USB-C Device: connect VBUS, D+/D−, GND, CC1/CC2(5.1k to GND). Optional ESD TBD.

J3 SWD 1=+3V3 2=SWDIO 3=SWCLK 4=GND

=== B. Regulated iso 5V + USB Suspend load switch ===

U2=Traco TMR 0511 REGULATED 2W 5V/400mA (4.5-9Vin). Backup: TMR 1-0511 (1W/200mA SIP-6). Suspend: ISO_PWR_EN=PA15 low cuts Q1.

=== C. READY path (wired) ===

READY: MCU high -> LED on -> VO low -> invert -> READY_ISO high -> enable all data VE. Default READY_ISO low.

r5.2: U13 unused inputs pin9/11/13 tied to GND_ISO (isolated side). Was GND (host) in r5.1 — expert review fix.

=== D. Eight data channels (wired) ===

--- Ch D0 / DB25 Pin 2 ---

--- Ch D1 / DB25 Pin 3 ---

--- Ch D2 / DB25 Pin 4 ---

--- Ch D3 / DB25 Pin 5 ---

--- Ch D4 / DB25 Pin 6 ---

--- Ch D5 / DB25 Pin 7 ---

--- Ch D6 / DB25 Pin 8 ---

--- Ch D7 / DB25 Pin 9 ---

Each ch (r6): VBUS->R1x->6N137 A, C->LED_Kx->LVC06 sink (input MARKER_Dx); VE=READY_ISO; VO+R2x; 74HC14; R3x->DB25_Dx; R4x to GND_ISO. U12=D0-5, U13=D6-7(+READY inv).

=== E. DB25 female ===

Pins 1,10–17: NC. Silk: TRIGGER IN ONLY / 禁止 Out(Amp)

=== Legend ===

U3–U10 data 6N137; U11 READY 6N137; U12 74HC14 D0–D5; U13 74HC14 D6–D7+READY; U2 iso; U14 3V3; U1 MCU; J1 USB-C; J2 DB25; J3 SWD

NOT PCB-fab frozen. Expert schematic review next.

=== F. r6 LED sink driver (expert r5 item 1; see 6N137驱动电流核算-r6.md) ===

6N137 LED: VBUS -> Rled 330R 1% -> A(pin2); C(pin3) -> LED_Kx -> SN74LVC06A open-drain sink (GND). MCU high = LED on (polarity unchanged).

IF worst min 6.15 mA (VBUS 4.40 V, VF 1.80 V, VOL 0.55 V, R+1%); max 13.8 mA (VBUS 5.50 V, VF 1.00 V). 47k pull-downs: LEDs off during reset/boot; Ioff: off when 3V3 absent.

R80–R88 47k: LVC06 input pull-downs (GPIO Hi-Z at reset -> output off -> LED off)

C23: U13 74HC14 decoupling (pin14 VCC_ISO – pin7 GND_ISO)

R89 = HSE Rext, fitted 0R. 776 ohm (XC of C6) fails gain margin (0.5) and is not a fit value. Ceiling if T9 drive > 200 uW: 22 ohm. See docs/补充审查确认-r7.1.md.

R72 4.7k: PA15 (JTDI) internal pull-up 30–50k at reset -> ISO_PWR_EN <= 0.45 V (Q2 off). R84 4.7k: PB4 (NJTRST) pull-up -> MARKER_D4 <= 0.45 V < LVC VIL 0.8 V.

Y1 NDK NX3225GA-25.000M-STD-CRG-2 (CL 8 pF, ESR 50 ohm max): C5=C6=8.2 pF (Cstray 4 pF). gm_crit mA/V < Gm_crit_max 1 mA/V.

=== r7.2 host boost + DB25 TVS (NOT FOR FAB, PCB not updated) ===

U20 TPS61085PWR: Vout=1.238*(1+74.5k/18.2k)=6.305 V. Q1 stays upstream of the inductor.

L201 XAL7030-682MEC 6.8 uH Isat 8.7 A. D201 SS34 anode at SW, cathode at ISO_VIN. U18/U19 TPD4E05U06DQAR to GND_ISO only.

r7.2 gate FIX: R70 33k to VBUS; R96 3.3k / C207 220nF slew Q1. R98 1k senses Q1_GATE into high-Z Q3 gate; NO R99 pulldown/divider. Q3 discharges ISO_VIN as Q1 opens OFF. PREVIOUS PCB/DRC STALE: reroute before fabrication.

NOT FOR FAB — r7.2 schematic and PCB, no Gerber and no powered test
r7.2 schematic: TPS61085PWR ~6.30 V boost after Q1, DB25 D0–D7 TVS to GND_ISO. Q3 gate fix: R98
marker-box
Sheet: /
File: usb_db25_marker.kicad_sch
Title: USB CDC to DB25 8-bit Marker Adapter
Size: A0 Date: 2026-10-19 **Rev: r7.2**
KiCad E.D.A. 9.0.9 Id: 1/1